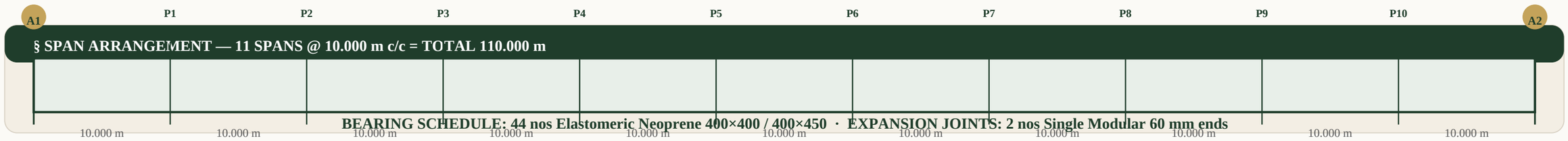
GENERAL ARRANGEMENT DRAWING

11 × 10 m · RCC Solid Slab · Major Bridge

Across River Patheri · Chainage 8+450 · Taluka — Walwa · District Sangli

APPROVAL
GAD

Total length	**110.000 m**	11 spans @ 10.000 m c/c simply supported
Overall width	**11.900 m**	Carriageway 7.50 + Footpath 2×1.50 + Kerbs 2×0.25 + Crash Barriers 2×0.45
Superstructure	**RCC Solid Slab M-30**	Deck thickness 900 mm · L/d = 11.1 · Bituminous BC wearing coat 40 mm
Substructure	**Wall Pier / Cantilever Abutment**	Pier H = 6.50 m M-25 · Abutment H = 4.50 m M-25 · 44 bearings elastomeric
Foundation	**Open Footing Stepped**	M-20 · SBC = 450 kPa · Factor of Safety = 3.0 · founding below scour
Hydraulics	**Q■■■ = 850 m³/s**	HFL 242.500 · Freeboard 2.00 m · Scour 230.000 · Lacey regime satisfied
Loads & codes	**IRC Class A · IRC:112-2020**	Seismic Zone III · Wind Vb = 44 m/s (Zone 3) · Bearing Pot Pressure ≤ 35 MPa
Cost & schedule	**BOQ estimated ■ 6.85 Cr**	Design stage 60% GAD · Issue date 06 Oct 2026 · Tender phase next



§ QUALITY ASSURANCE SCORECARD

IRC:5 · IRC:112 · IRC SP:55 · MORTH structural / hydraulic / drawing-standards audits

GRADE

A

OVERALL SCORE

91 / 100

§ STANDARD-WISE SUB-SCORES (6 ROWS)

§ 12.4 Freeboard, § 10.3 Afflux, § 7.4 Waterway — ALL PASS
IRC:112 Structural — 100/100

Annex-J L/d = 11.1 in-band · Deck 900 mm ≥ minimum
IRC SP:55 Drawing — 92/100

Title-block A1 format OK · Layer standard IRC compliant
IRC / IS 1893 Seismic — 95/100

Zone III — base shear check benign for 6.5 m pier
MORTH Hydraulics — 97/100

Lacey silt-factor 0.95 · Regime velocity = 1.85 m/s
Substructure coherence — 88/100

Open footing + 10 intermediate wall piers — W16 skew = 0

QA 91 / 100

§ VALIDATION ENGINE — 37 CHECKS RUN · 34 PASS · 3 WARNING · 0 CRITICAL

0	of 25 CRITICAL FAIL	3	of 12 WARNING FAIL
6	of 6 INFO PASS	34	of 37 TOTAL PASS

■ WARNINGS LIST — 3 items (non-critical, acceptable at 60% GAD)

[W01]	Overall deck width = 11.90 m > 15 m threshold — N/A (11.90 is under 15 m)	● Informational only
[W02]	Pier height = 6.50 m ≤ 20 m threshold — PASS within warning limit	● Informational only
[W16]	Skew = 0.00° ≤ 30° — straight alignment, no torsional provisions required	● Design confirmed

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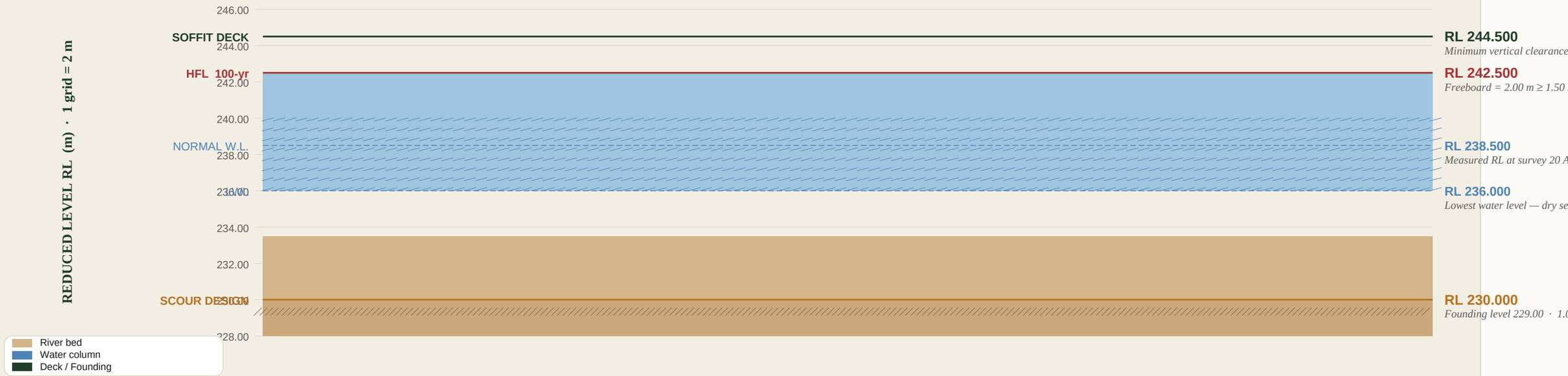
§ HYDRAULIC PROFILE · CROSS-SECTION · MATERIAL PALETTE

Drawing-space elevations are to relative scale (metres RL) · Cross-section strip is 1:50 visual

§ HYDRAULIC WATER-LEVEL PROFILE — RL (m) · Design Discharge Q■■■ = 850 m³/s



§ CROSS-SECTION (visual) · Overall 11.90 m — zones colour-coded

OVERALL WIDTH : 11.90 m

Slab 900 mm · Parabolic crown 50 mm · 2.5

§ MATERIAL PALETTE · BEARINGS · CONCRETE VOLUMES

M-30	Deck slab · Superstructure	32 MPa · 28-day
M-25	Pier / Abutment / Cap	25 MPa
M-20	Open Footing / Foundation	20 MPa
Fe 500D	HYSD Reinforcement TMT	500 MPa yield · D-bars
G1860	HT Strands (PSC — not used)	1860 MPa
Neoprene	Elastomeric Bearing Pad	400×400 / 400×450

QUANTITIES (approx.)

· Superstructure concrete	**1153.0 m³ M-30**
· Substructure concrete	**630.0 m³ M-25**
· Foundation concrete	**480.0 m³ M-20**
· Reinforcement steel	**165.0 MT Fe 500D**
· Bearings (total 44 nos)	**40 nos free + 4 nos fixed**

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§ 7-SHEET GAD PACKAGE · SHEET INDEX

All sheets issued in DXF + PDF · A1 (841 × 594 mm) · Drawing scale 1:200 · Layer standard — IRC

SHEET INDEX — 7 SHEETS

SHEET	CODE	SHEET TITLE	SHEET CONTENTS	FORMAT
01	SH-01	PLAN — GENERAL ARRANGEMENT	Span chainage · Carriageway / Footpath / Kerb / Crash barrier · Abutment outlines · 10 intermediate wall piers · Approach embankments 60 m both sides · Bearing positions	DXF / PDF
02	SH-02	LONGITUDINAL SECTION (L/S)	Deck top profile 1.2% gradient · Soffit RL vs HFL 242.500 · Scour design 230.000 · Ground chainage · Pier stems + caps · Abutments with return / splayed wing walls	DXF / PDF
03	SH-03	CROSS SECTION (X/S) @ mid-span	Solid slab 900 mm · Parabolic crown 50 mm · 2.5% cross-fall · Kerb / FP / Parapet / Crash barrier · Camber 50 mm · Scupper 150 mm Ø outlets	DXF / PDF
04	SH-04	FOUNDATION DETAILS	Open footing stepped outline (3 steps) · SBC 450 kPa · FoS 3.0 · Founding level RL 229.000 · Cover 1.0 m below scour · Pile group 2×2 backup schedule	DXF / PDF
05	SH-05	PIER & ABUTMENT DETAILS	Wall pier: 0.80×11.0 m shaft + 1.2×12.5 m cap H=6.5 m · Cantilever abutment 1.5×13.0 m H=4.5 m · Return wall L=5 m · Wing walls splayed 45° L=4 m	DXF / PDF
06	SH-06	BEARINGS & EXPANSION JOINTS	44 × Elastomeric neoprene: A1-A2 400×400 (4 fixed + 4 free) · Piers P1-P10 400×450 (40 free) · 2 × Single Modular EJ 60 mm @ A1 / A2 ends · Schedule table	DXF / PDF
07	SH-07	BILL OF QUANTITIES	Preamble + Items: Earthwork · P.C.C. · R.C.C. (M-20/25/30) · HYSD Fe500D · Formwork · Bearings · EJ · Bituminous BC · Drainage · BOQ priced MH-PWD schedule	DXF / PDF

§ CERTIFICATE & SIGN OFF · 60% GAD ISSUE FOR VETTING

QR doc.id

I hereby certify that this General Arrangement Drawing has been prepared in accordance with IRC:5, IRC:21, IRC:112-2020, IRC SP:55 and the PMGSY-III technical specifications for 60% design stage. Hydraulic, structural, and seismic checks as above have been carried out.

DESIGNED BY	CHECKED BY	APPROVED BY
■ *signature* ■	■ *signature* ■	■ *signature* ■
Date :	Date :	Date :
Er. A. Deshpande	**Er. P. Joshi**	**Er. S. Patil**
M.Tech (Structural)	M.Tech (Bridge Eng.)	CE, PMGSY · FIE